Barcelona School of Economics
SUMMER SCHOOL

2023

# Firms in Macro-Finance: Tools and Applications

## Professor(s)

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## Overview and Objectives

Firm behavior– the process of entry, innovation, and growth – shapes aggregate productivity, employment, and output. This course presents a rigorous analysis on the mapping from firm-level frictions and shocks, to firm decisions, to industry-level and aggregate outcomes. We will illustrate the analytical tools necessary to solve and analyze firm dynamics models with frictions and misallocation, and the empirical tools to test their predictions.

A special emphasis will be placed on financial frictions, and the course will answer questions such as: How do we detect firm-level financial frictions? What are their real effects? Are they important to explain cross-country differences in productivity and output? Do financial shocks drive business cycles? Do financial frictions amplify the cyclical effects of other shocks?

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## Course Outline

- Session 1 (Monday)
  - Theory: stylised facts on firm dynamics and introduction to misallocation
  - Practical session: Introducing Stata code to compute productivity and misallocation
- Session 2 (Tuesday)
  - Theory: dispersion in productivity, frictions, and misallocation of resources: theoretical framework and empirical evidence.
  - Practical session: Stata code on misallocation, introducing Matlab code on firm dynamics.
- Session 3 (Wednesday)
  - Theory: micro foundation and empirical testing of firm-level financial frictions, and of their aggregate implications.
  - Practical session: Stata code on financial frictions and misallocation.
- Session 4 (Wednesday-Thursday)
  - Theory: firm dynamics and financial frictions: model and applications.
  - Practical session: Matlab code on firm dynamics with financial frictions.
- Session 5 (Thursday July 7th and Friday July 8th)
  - Theory: finance, firm dynamics and the business cycle.
  - Practical session: Matlab code on firm dynamics with financial frictions.

## Required Activities

Lecture notes will be provided before each class. Below, references are separately listed for each session.

**Papers denoted with * will be discussed in the lectures. Papers denoted with ** will be covered with some depth in the lectures.** having a look at these papers in advance might be useful to better follow the lectures.

Selected tools and applications (both simulating firm dynamics models as well as some empirical applications) will be illustrated in detail during practice sessions.

## List of References

**1) Stylised Facts on Firm dynamics**


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**2) Dispersion in productivity, frictions, and misallocation of resources: theoretical framework and empirical evidence**


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**3) Micro foundation and empirical testing of firm-level financial frictions, and of their aggregate implications.**


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